

05 CONTINUOUS RANDOM VARIABLES (notes p112-p147)

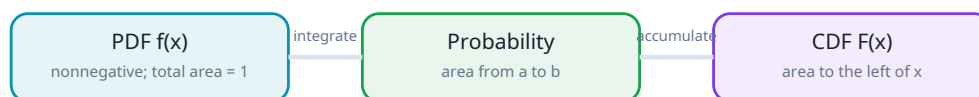
1. Why continuous needs new machinery (p113-p116)

- T = time of peak electricity demand at a power plant
- cannot be limited to a countable list of times
 - T can take ANY value in $[0, 24)$
 - $P(\text{peak at exactly } 12.013278...) = 0$ in a continuous model

A random variable is CONTINUOUS if it can assume any value in some interval, AND the probability of any specific single value is 0.

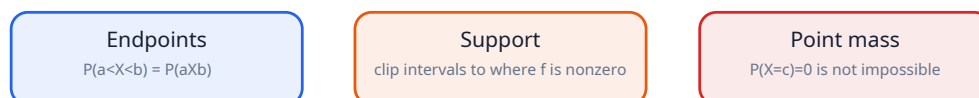
2. The probability density function, pdf (p117)

Continuous probability: area, accumulation, derivative



$$F(x) = \text{integral from } -\infty \text{ to } x \text{ of } f(t) \, dt$$

$f(x) = F'(x)$, where an absolutely continuous F is differentiable



continuous interval: $F(b) - F(a)$

integer-valued interval: $F(b) - F(a-1)$

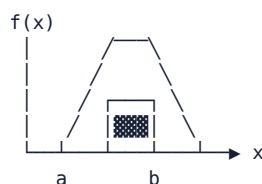
PDF, probability area, and CDF relationships

$f(x)$ is a pdf if:

- $f(x) \geq 0$
- $\int_{-\infty}^{+\infty} f(x) \, dx = 1$
- $P(a \leq X \leq b) = \int_a^b f(x) \, dx$

$f(x)$ IS A DENSITY, NOT A PROBABILITY.
 $f(x)$ can be GREATER THAN 1.
 e.g. uniform on width 0.5 $\rightarrow f = 1/0.5 = 2$
 only its INTEGRAL is a probability.

$P(a \leq X \leq b) = \text{AREA under the density curve between } a \text{ and } b$:



the shaded AREA = $P(a \leq X \leq b)$

3. The five remarks (p119-p123) - each is a possible true/false

- R1 if $\int f = k$ and $k \neq 1$, then $f(x)/k$ IS the pdf
(this is how you "find k" - normalise)
- R2 X takes all values in (a,b) , where a,b may be $-\infty, +\infty$
- R3 $P[X = x_0] = \int \text{from } x_0 \text{ to } x_0 f \, dx = 0$ (zero width, zero area)
- R4 probability zero does NOT mean impossible for continuous rv
 $P(A) = 0$ does NOT imply $A = \emptyset$ <- subtle, examinable
- R5 if X only lives in $[a,b]$, set $f(x) = 0$ for all x outside $[a,b]$
(the SUPPORT convention)

R3 + R4 TOGETHER:
 $P(X = c) = 0$ for every point c
 yet SOME point must occur
 so "probability 0" and "impossible" are DIFFERENT here
 and $P(a < X < b) = P(a \leq X \leq b)$ exactly (endpoints free)

4. Worked: $f(x)=2x$ (p124-p125)

$f(x) = 2x$, $0 < x < 1$, 0 elsewhere

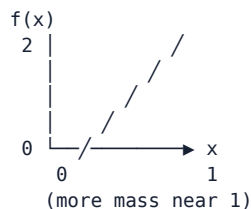
CHECK it is a pdf:

$$f(x) \geq 0 \quad \checkmark$$

$$\int_{-\infty}^{\infty} f = \int_0^1 2x \, dx = [x^2]_0^1 = 1 \quad \checkmark$$

$$P[X \leq 1/2] = \int_0^{1/2} 2x \, dx = [x^2]_0^{1/2} = 1/4$$

the density $f(x)=2x$ rises linearly:



5. Worked: continuous conditional (p126)

$$P[X \leq 1/2 \mid 1/3 \leq X \leq 2/3]$$

$$= \frac{P[1/3 \leq X \leq 1/2]}{P[1/3 \leq X \leq 2/3]} = \frac{\int_{1/3}^{1/2} 2x \, dx}{\int_{1/3}^{2/3} 2x \, dx}$$

$$= \frac{(1/4 - 1/9)}{(4/9 - 1/9)} = \frac{(5/36)}{(1/3)} = 5/12$$

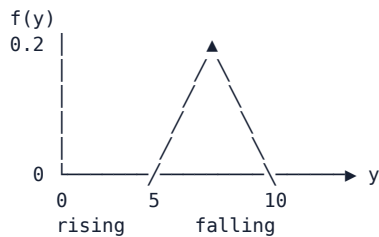
CONDITIONAL IN THE CONTINUOUS WORLD: fraction of AREAS.
 Same rule $P(A|B) = P(A \cap B) / P(B)$, but P 's are integrals.
 This exact shape is re-asked on the ETE S4 paper.

6. Worked: bus waiting time, Y (p127-p129) - the 6-part monster

two buses, each wait $\sim U(0,5)$, total wait Y has pdf:

$$f(y) = \begin{cases} y/25 & 0 \leq y < 5 \\ 2/5 - y/25 & 5 \leq y \leq 10 \\ 0 & \text{otherwise} \end{cases}$$

the triangular density:



(b) VERIFY total area = 1:

$$\begin{aligned} & \int_0^5 y/25 \, dy + \int_5^{10} (2/5 - y/25) \, dy \\ &= [y^2/50]_0^5 + [2y/5 - y^2/50]_5^{10} \\ &= (25/50) + [(4 - 2) - (2 - 1/2)] \\ &= 1/2 + [2 - 1.5] = 1/2 + 1/2 = 1 \quad \checkmark \end{aligned}$$

$$(c) P(Y \leq 3) = \int_0^3 y/25 \, dy = [y^2/50]_0^3 = 9/50 = 0.18$$

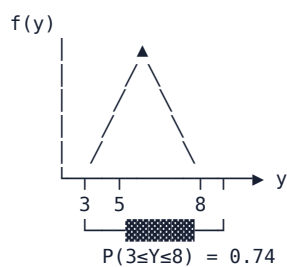
$$\begin{aligned} (d) P(Y \leq 8) &= \int_0^5 y/25 \, dy + \int_5^8 (2/5 - y/25) \, dy \\ &= 1/2 + [2y/5 - y^2/50]_5^8 \\ &= 1/2 + [(3.2 - 1.28) - (2 - 0.5)] \\ &= 1/2 + [1.92 - 1.5] = 1/2 + 0.42 = 0.92 \end{aligned}$$

$$(e) P(3 \leq Y \leq 8) = P(Y \leq 8) - P(Y < 3) = 46/50 - 9/50 = 37/50 = 0.74$$

$$\begin{aligned} (f) P(Y < 2 \text{ or } Y > 6) &= \int_0^2 y/25 \, dy + \int_6^{10} (2/5 - y/25) \, dy \\ &= 4/50 + 32/50 = 0.08 + 0.32 = 0.40 = 2/5 \end{aligned}$$

ERRATA 13: the slide's working for (f) prints the first integral bound as 3, not 2, which contradicts its own stated event " $Y < 2$ ". The final answer $2/5 = 0.40$ is still correct for " $Y < 2$ or $Y > 6$ ". Use $0.08 + 0.32 = 0.40$.

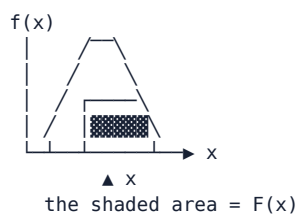
the (e) picture:



7. The cdf for continuous (p131-p134)

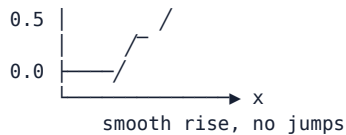
$$F(x) = P(X \leq x) = \int_{-\infty}^x f(y) \, dy$$

$F(x)$ = the AREA to the LEFT of x



$F(x)$ is a smooth S-curve (not a staircase like the discrete cdf):





8. Using F(x) (p134-p135)

$$P(X > a) = 1 - F(a)$$

$$P(a \leq X \leq b) = F(b) - F(a)$$

CONTINUOUS vs INTEGER: the SAME theorem has two forms

continuous : $P(a \leq X \leq b) = F(b) - F(a)$

integer rv : $P(a \leq X \leq b) = F(b) - F(a-1)$ <- the a-1

why the a-1:

continuous: $F(a)$ excludes the zero-width point a -> fine

integer : $F(a)$ INCLUDES the value a -> must drop it

9. Getting f from F (p136-p137)

for discrete : pmf = DIFFERENCE of two cdf values

for an absolutely continuous distribution: pdf = DERIVATIVE of the cdf

$$F'(x) = f(x) \quad \text{where } F \text{ is differentiable}$$

10. Worked: find the cdf, exercise 1 (p138)

$$(i) f(x) = \begin{cases} 1/3 & 0 \leq x \leq 1 \\ 2/3 & 1 < x \leq 2 \\ 0 & \text{elsewhere} \end{cases}$$

$$\text{CHECK: } \int_{-\infty}^{\infty} f(x) dx = \int_0^1 1/3 dx + \int_1^2 2/3 dx = 1/3 + 2/3 = 1 \quad \checkmark$$

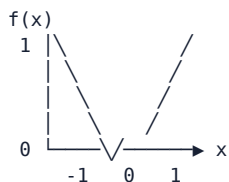
$$F(x) = \begin{cases} 0 & x < 0 \\ x/3 & 0 \leq x \leq 1 & (\int_{-\infty}^x 1/3 du = x/3) \\ 1/3 + 2(x-1)/3 & 1 < x \leq 2 & (1/3 + \int_1^x 2/3 du) \\ 1 & x > 2 \end{cases}$$

$$(ii) f(x) = |x|, \quad -1 < x < 1$$

split at 0 because of the absolute value:

$$F(x) = \begin{cases} 0 & x \leq -1 \\ \int_{-1}^x (-u) du & -1 < x \leq 0 & = (1 - x^2)/2 \\ \int_{-1}^0 (-u) du + \int_0^x u du & 0 < x < 1 & = (1 + x^2)/2 \\ 1 & x \geq 1 \end{cases}$$

the $|x|$ density is a V:



11. Worked: cdf to pdf (p139) - the reverse direction

$$F(x) = \begin{cases} 0 & x < 1 \\ (x-1)^4 / 16 & 1 \leq x \leq 3 \\ 1 & x > 3 \end{cases}$$

(i) DENSITY = derivative:

$$f(x) = d/dx [(x-1)^4 / 16] = 4(x-1)^3 / 16 = (x-1)^3 / 4, \quad 1 \leq x \leq 3$$

0 elsewhere

$$(ii) P(2 \leq X \leq 3) = F(3) - F(2)$$

$$= 1 - (2-1)^{4/16}$$

$$= 1 - 1/16 = 15/16$$

USING F IS THE FAST ROUTE. If the question gives F, do NOT integrate the density you just found. Just subtract two F values.

12. Expected values, continuous (p140-p144)

summation $\rightarrow$ integration
pmf $\rightarrow$ pdf

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx$$

$$E[h(X)] = \int_{-\infty}^{\infty} h(x) f(x) dx$$

$$E(aX+b) = a E(X) + b \quad (\text{same linear rule})$$

$$\text{Var}(X) = \int (x-\mu)^2 f(x) dx = E(X^2) - [E(X)]^2$$

$$\sigma = \sqrt{\text{Var}(X)}$$

13. Worked: hospitalization, $Y=X+4$ (p142) - and its ETE sibling

$Y = X + 4$, where $f(x) = 32/(x+4)^3$, $x > 0$
find $E(Y)$

$$E(Y) = E(X+4) = E(X) + 4$$

$$E(X) = \int_0^{\infty} x \cdot 32/(x+4)^3 dx$$

substitute $u = x+4$, $x = u-4$, $dx = du$, limits $4 \rightarrow \infty$:

$$= 32 \int_4^{\infty} (u-4)/u^3 du$$

$$= 32 \int_4^{\infty} (u^{-2} - 4u^{-3}) du$$

$$= 32 [-u^{-1} + 2u^{-2}]_4^{\infty}$$

$$= 32 [0 - (-1/4 + 2/16)]$$

$$= 32 [1/4 - 1/8] = 32 (1/8) = 4$$

$$E(Y) = 4 + 4 = 8 \text{ days}$$

THE TRICK: $Y = X + 4$, so use the SHIFT rule.
Do NOT integrate $(x+4)$ against the density.
 $E(X+4) = E(X) + 4$ is one line.

14. Worked: Pareto density (p146-p147)

$$f(x) = k/x^{(k+1)}, \quad x \geq 1, \quad k > 0$$

$$\text{normalise: } \int_1^{\infty} k x^{-(k+1)} dx = k [-x^{-k}/k]_1^{\infty} = 1 \quad \checkmark$$

$E(X)$ exists only if $k > 1$ (otherwise the integral diverges)
 Var exists only if $k > 2$

THE CONVERGENCE CONDITION IS EXAMINABLE.
A heavy tail can make $E(X)$ or $\text{Var}(X)$ INFINITE.
Always check the exponent condition before claiming a mean.

15. The continuous toolkit, one block

check a pdf	$\int \text{over support} = 1$
find k	set up that integral = 1, solve
$P(a \leq X \leq b)$	$\int_a^b f$, or $F(b) - F(a)$
cdf	$\int_{-\infty}^x f$
pdf from cdf	$f = F'$

E(X)	$\int x f(x) dx$	
Var(X)	$E(X^2) - [E(X)]^2$	
conditional	ratio of two integrals	

That is the whole continuous part. Next: the two named discrete distributions.